SEEING THE UNSEEABLE

Instruments help scientists gather information about amazing things that we cannot see, hear, smell, taste, or feel with our senses. Thanks to these tools and techniques, we can peer inside planets and deep into space. We can observe particles too tiny to see and even look back in time!

Parker Solar Probe

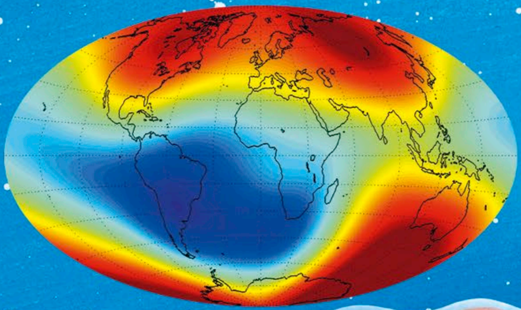
Space probes contain lots of instruments for exploring places too difficult to visit in person. This rocket sent the Parker Solar Probe hurtling deep into the scorching atmosphere of the sun itself.

Atmospheric radar dish

By bouncing radio waves off an object, radar systems can tell us its shape, speed, and direction of travel. This huge dish is part of a radar system that monitors storms and other weather in the Earth's atmosphere.

Magnetic field

Geophysicists use instruments to measure the Earth's magnetic field. It is not steady, but varies across space and time. Understanding it is important for navigation, plotting maps, and detecting minerals deep under the ground.



Ice core

The deepest layers of ice that cover Antarctica and Greenland are hundreds of thousands of years old. By drilling down and taking pieces out of this ice, scientists can obtain trapped bubbles of ancient air. This tells us what the climate was like in the past.